

Machine Learning 101: Housing Price Prediction

This project is designed to introduce the core pipeline of machine learning using Linear Regression. The goal isn't just to write code that runs, but to understand the relationship between data inputs and predictive outputs.

The Dataset

Use the **California Housing Dataset** (which can be loaded directly through Scikit-Learn's datasets module) or the **Ames Housing Dataset** from Kaggle. These datasets contain features like median income, house age, total rooms, and location to predict the median house value.

The Mission: Step-by-Step

Step 1: Data Acquisition & Setup

Set up a Python environment (Jupyter Notebook is highly recommended for beginners). Import the standard data science stack: `pandas`, `numpy`, `matplotlib.pyplot`, and `sklearn`. Load the dataset into a Pandas DataFrame and print the first 5 rows to see what you're working with.

Step 2: Exploratory Data Analysis (EDA)

Before throwing algorithms at the wall, look at the data. Plot a histogram of the target variable (house prices). Create a scatter plot comparing a single feature—like median income or square footage—against the house price. Do you see a correlation? Is the relationship linear?

Step 3: Data Preprocessing

Real-world data is rarely perfect. Check the DataFrame for missing values (NaNs) and decide how to handle them (e.g., dropping them or filling them with the median). Next, split the dataset into a **Training Set** (80%) and a **Testing Set** (20%) using `train_test_split`.

Crucial concept to grasp here: Why is it a massive mistake to test the model on the exact same data you used to train it?

Step 4: Training the Model

Initialize a `LinearRegression` model from Scikit-Learn. Fit the model to your training data (`X_train` and `y_train`). Start with **Simple Linear Regression** (using just one input feature, like income) before upgrading to **Multiple Linear Regression** (passing in all available features).

Step 5: Evaluation & Metrics

Use your trained model to predict prices on your unseen testing set (`X_test`). Calculate the **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)**. Translate these metrics from math into reality: What do these specific numbers mean in plain English in the context of dollars?

THE "PROVE IT" CHALLENGE

Don't let her stop at just printing out an error number. To actually show she understands how the model thinks, complete these two tasks:

- **Visualize the Error:** Plot the actual testing prices vs. the predicted prices on a scatter plot. A perfect model forms a perfectly straight diagonal line. How scattered are the dots around that ideal line?
- **Interpret the Math:** Extract the `.coef_` (coefficients) from the trained model. Which feature does the model think has the biggest impact on the housing price? If a house adds one more bedroom, exactly how much money does the algorithm say the price will increase?